

Mitigating Stillbirth Risk Using Machine Learning

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Dataset: [https://www.kaggle.com/datasets/chrisheiner/
national-vital-stastitics-fetal-death-2016-2023](https://www.kaggle.com/datasets/chrisheiner/national-vital-stastitics-fetal-death-2016-2023)

Code: https://github.com/cheiner-27/fetal_death_prediction

Abstract

This project investigates the feasibility of developing a machine learning algorithm to mitigate stillbirth risk by leveraging the CDC's Division of Vital Statistics birth and fetal death datasets. With over 20,000 fetuses lost annually after the 20th week of pregnancy, many of these deaths appear to be preventable through systematic monitoring and timely clinical intervention. While individual risk factors such as chronic hypertension are well understood, the complex intersection of multiple maternal and fetal comorbidities presents a high-dimensional problem that necessitates the use of advanced predictive modeling. The strategy involved utilizing large-scale National Vital Statistics System (NVSS) data to train and evaluate multiple machine learning models, translating vast patient data points into structured risk levels that could provide clinicians with a tool for more informed decision-making. The results suggest that a predictive model can highlight previously unexplored variable interactions contributing to fetal mortality. Although the dataset presents limitations, the findings underscore the potential for machine learning to enhance trust and transparency between providers and patients.

Contents

Introduction	4
Problem Description	4
Motivation	4
Related Work	6
Data	7
Dataset Description	7
Data Analytics Tools	8
Data Profile and Preprocessing	8
Descriptive Statistics	9
Exploratory Data Analysis	9
Figure 1: Birth Weight and Gestational Age Distribution	10
Figure 2: Risk Factor Prevalence	11
Figure 3: Standardized Mean Differences	12
Predictive Models	13
Logistic Regression	13
CatBoost	14
AutoGluon Ensemble	15
Model Comparison	16
Results and Discussion	17
Findings	17
Performance Summary	21
Ethical Considerations	23

Privacy	23
Explainability	23
Algorithmic Bias	24
Conclusion and Future Work	24
Conclusion	24
Recommendations for Future Work	25

Introduction

Problem Description

Each year, over 20,000 fetuses die after the 20th week of pregnancy (Gregory et al., 2023). In many instances, fetal death is preventable through proper monitoring and timely intervention. Monitoring can include additional scans, blood tests, urine analysis, growth tracking, genetic evaluation, and fetal assessment using a fetoscope or Doppler (Johns Hopkins Medicine, n.d.). When abnormal results are detected, a physician can intervene by prescribing medication, transfusing blood, or performing minimally invasive procedures and open fetal surgery (Lin et al., 2021). It is critically important that obstetricians determine the correct level of monitoring and intervention; while insufficient care may result in an adverse outcome, excessive intervention adds unnecessary strain on the healthcare system.

Each medical diagnosis a mother receives during pregnancy can alter the probability of a successful delivery. While individual risk factors such as chronic hypertension or preeclampsia are well understood in isolation, the way these comorbidities intersect and interact with other maternal and fetal characteristics has not been deeply explored (Wolfson et al., 2025). Effectively assigning a risk level while accounting for these numerous, complex interactions presents an exponentially scaling problem. This complex multivariate environment makes machine learning an ideal application, as algorithms can efficiently ingest the vast number of data points collected for each patient and translate them into a structured risk level or probability of fetal death. The research problem examined here is whether such an algorithm can be constructed using the CDC’s Division of Vital Statistics birth and fetal death datasets.

Motivation

For most expectant families, the loss of a pregnancy is the greatest tragedy they can experience. The ratio of live births to stillbirths approaches 90:1, making the event unexpected, particularly for mothers with no notable comorbidities. The specific cause of

death can vary in terms of clinical understanding, as many risk factors associated with fetal death are present in women with uncomplicated pregnancies (Silver, 2007). Parents experiencing a stillbirth are significantly more likely to develop a mental health disorder than parents with live births (Westby et al., 2021). As someone who has personally experienced a pregnancy loss and who has developed data analysis skills in both academic and professional settings, I felt a moral obligation to attempt to develop a model that helps address this problem.

Beyond the immediate emotional impact, predictive models for fetal risk offer significant advantages to both clinical practitioners and the public health sector. Fetal mortality is a frequently cited statistic for assessing the quality of a regional healthcare system, with the WHO identifying its reduction as a core Sustainable Development Goal (WHO, n.d.). A sufficiently accurate model can be used by providers for more informed and systematic decision-making. In addition to assessing a pregnancy’s baseline risk level, clinicians could estimate the impact of specific procedures on pregnancy metrics to observe how the model evaluates the downstream effect on overall fetal risk. Even a less accurate model that cannot be used in a clinical setting may still be of value—such a model may highlight unexplored variable interactions that warrant further investigation. These investigations are needed because a Swedish survey conducted in 2019 found that nearly half of women were given no explanation as to why their babies died (Höglund et al., 2019). Improvements in this area not only enhance patient outcomes but also improve trust and transparency between patients and their providers.

Outside of the more immediate impacts, this project has the potential to benefit the machine learning and data science community. Because NVSS data is only captured at the time of the event, the current data structure does not allow for a truly dynamic model that updates a patient’s prediction over time. By increasing the medical community’s familiarity with machine learning, the need for large volumes of high-quality data becomes more apparent. This awareness may eventually lead to the public release of Electronic Health

Record (EHR) data scrubbed of Personally Identifiable Information (PII). Such datasets would be immensely valuable to the community, as predictive models could be developed for problems that cannot be solved by currently available data.

Related Work

The highly regulated nature of the medical industry, particularly in sensitive areas like obstetrics, leads to slower adoption of new technologies and less commercial interest. These demand factors have upstream impacts on the amount of academic resources dedicated to the topic. As discussed by Cersonsky et al. (2023), many papers have models with accuracy unsuitable for a clinical environment. In other instances where models have better accuracy, they are not practically applicable as they utilize variables that would not be known ahead of time, such as the cause of delivery or hospital of record. Industry adoption of tools utilizing the technology is limited as well, with only four devices on the FDA’s Artificial Intelligence-Enabled Medical Devices list with obstetrics and gynecology panel leads (FDA, 2025). Taken together, this research area is still in its early stages and there are multiple areas for potential improvement.

Building on the gaps identified in prior work, this project focuses on a few specific areas where improvements appear achievable without requiring clinical-only data. Cersonsky et al. (2023) produced a strong baseline model, but the study did not explore ensemble approaches. Ensemble models often sacrifice speed in exchange for better predictive performance, a tradeoff acceptable in this use case since predictions do not need to be generated in real time. Additionally, multiple studies noted that clinical decision-making during pregnancy—such as the choice to implement additional monitoring or earlier intervention—may influence outcomes and complicate retrospective modeling (Ferro de Mello et al., 2025; Cersonsky et al., 2023). These considerations shape the approach for this work, with the intent to reduce avoidable limitations seen in existing research so that the primary remaining constraint is the quality and structure of the underlying data.

Data

Dataset Description

The core datasets come from the CDC’s NVSS. Fetal death records span 1982 to 2023 and natality records span 1968 to 2024. Each fetal death file contains approximately 40,000 records with 140 features covering maternal demographic and pregnancy history characteristics, prenatal care indicators, gestational age and growth measures, and fetal death outcome descriptors. Due to variability in reporting requirements, many records contain imputed values, with separate indicator features flagging imputation. Most variables are coded and require consulting the PDF user guide associated with each file. The natality files are considerably larger—the 2024 file alone contains over 3.6 million records and 280 features—and additionally capture variables related to labor and delivery, neonatal outcomes at birth, and paternal information (NCHS, n.d.).

For this study, records spanning 2016 to 2023 were used. To address the extreme class imbalance, a random sample of 350,000 records was drawn from each annual natality file. Features not present across all files were removed, and a fuzzy match was performed to identify and standardize fields with slightly different names across years. The outcome of the pregnancy and the year of record were added as features, yielding 75 total features for exploratory analysis. Final feature counts for modeling depended on data cleaning and feature engineering strategies, though an 80/20 train-validation split was applied throughout.

Supplemental geographic social vulnerability data from the Agency for Toxic Substances and Disease Registry (ATSDR) (ATSDR, 2024) was initially intended to be incorporated. However, the publicly available NVSS files with geographic identifiers are approximately 1% of the size of the full dataset. The substantial reduction in scale would likely have restricted the model’s ability to generalize, so the decision was made to place greater emphasis on feature engineering from the core dataset instead.

Data Analytics Tools

Three distinct approaches to data extraction were required. Raw text files were parsed using a custom script that translated the content into a tabular format by reading field positions from user guide PDFs to define column boundaries. Translating data value mappings from PDFs was more involved because the table structures are inconsistent, often containing nested sub-rows for categorical data or misaligned columns. Google Gemini was utilized to create a specialized extraction tool for handling these structural complexities (Google, 2026).

Once processed, all data was loaded into Python using the pandas library. Exploratory visualizations were created with seaborn and matplotlib; final report visuals were produced in Tableau. Modeling was conducted in three stages. Initial architecture exploration was performed using AutoGluon, an AutoML framework that trains multiple models simultaneously and selects the best performers. Those architectures were then refined manually using XGBoost, scikit-learn, and Keras. A Git repository was maintained throughout for version control and reproducibility.

Data Profile and Preprocessing

A general data profile is provided in the Appendix (Tables 1–5), categorized by variable groupings. Two definitions are essential for interpreting these tables: *recode* variables represent existing data grouped into new categories, while *reporting flags* indicate whether a variable’s presence depends on the pregnancy’s geographic location. While the dataset is largely clean, some features contain mixed data types—for example, the “interval since last birth” uses 004–300 for months but uses 888 and 999 as sentinel values for “not applicable” and “unknown,” respectively. For modeling, these variables were split into explicit missing indicators to avoid treating “unknown” as a meaningful level.

Outliers in numeric variables were identified using the IQR method; categorical outliers were defined as those appearing in fewer than 1% of total records. The most critical refinement involved identifying potential data leakage—variables containing information not

available prior to labor were retained for visualization but removed before modeling to ensure predictive integrity.

Descriptive Statistics

The large number of features makes a comprehensive review impractical in this narrative; however, several features warrant particular attention. Across the dataset, an average gestational age of 38 weeks was observed, with the 75th percentile at 40 weeks. This aligns with research showing that a large proportion of labors are induced and often done so early (Kjerulff et al., 2017). The mean gestational age is one week below the median, reflecting downward pressure from the extreme low values present in fetal deaths.

Maternal age is another essential risk factor, as the risk of numerous complications increases with age (Hochler et al., 2023). The mean and median maternal age in the dataset are both 29, with a standard deviation of 5.91 years. Despite this central tendency suggesting lower age-related risk, the dataset contains 35,302 instances of maternal age below 18 and 24,976 instances above 43. The latter group is correlated with a significant increase in NICU admissions (Hochler et al., 2023).

The interval since last live birth contains mixed data types: over a third of records indicate a first pregnancy, while the remaining records show high variability, with an average interval of 50.72 months ($SD = 38.36$). The distribution has an extremely long tail, as evidenced by a maximum value of 300 months (25 years).

Exploratory Data Analysis

All visualizations were filtered to records with gestational age greater than 20 weeks to align with the fetal death definition. Because the natality files were sampled, visualizations relying on the relative proportion of fetal deaths to live births would overrepresent fetal deaths by approximately a factor of 10. This was carefully considered to ensure all visuals remained truthful and representative (Few, 2017).

Figure 1

Birth Weight Divergence: Fetal Deaths vs. Live Births (Weeks 20–47)

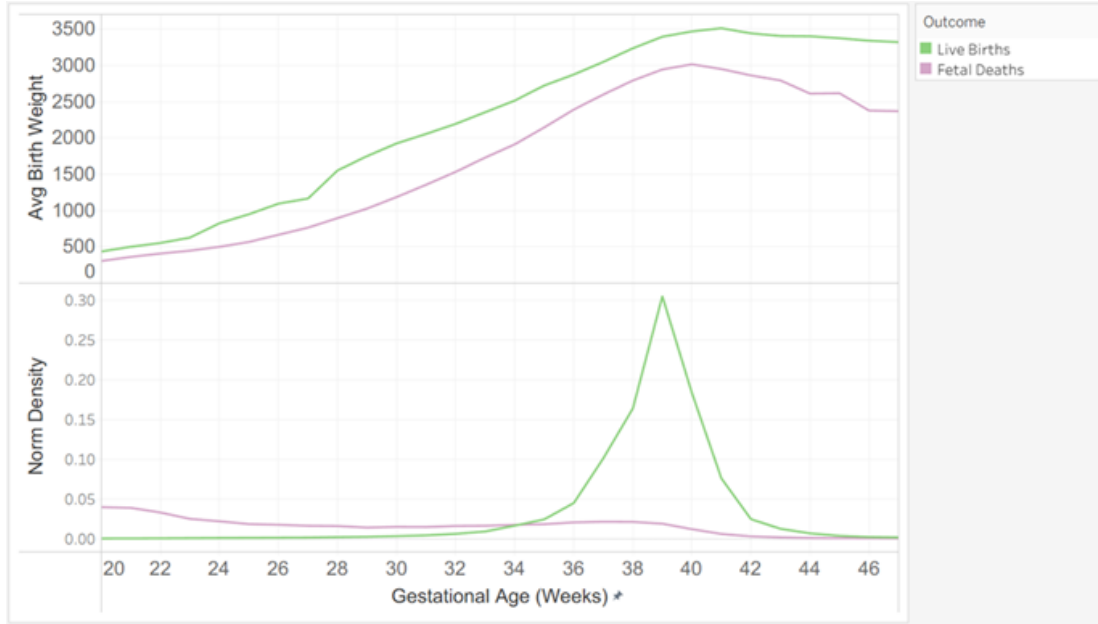


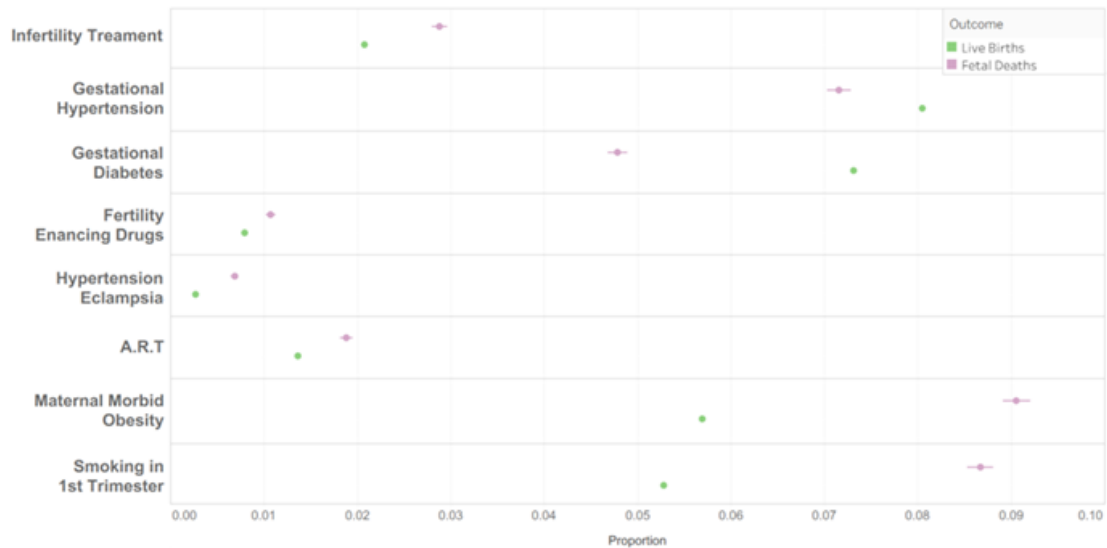
Figure 1: Birth Weight and Gestational Age Distribution

The first visualization is a combination of two charts. The top panel is a line chart displaying the average birth weight of live births and fetal deaths by gestational age. Birth weight for live births increases rapidly up to week 28 before following a linear trend through week 41, ultimately decreasing for later post-term births. The bottom panel is a probability density function of the two outcomes. Live births follow a mostly normal distribution centered at 39 weeks, while the distribution of gestational age for fetal deaths is comparatively flat, with slightly elevated frequency before 26 weeks.

The decrease in birth weight at post-term gestations likely reflects selection bias introduced by modern obstetric practices. Many hospitals induce labor when fetal macrosomia (birth weight above 4,000 g) is suspected, removing the heaviest fetuses from the pool of ongoing pregnancies and causing the average weight of remaining post-term births to appear artificially lower (Boulvain & Thornton, 2023). The steeper weight drop-off in fetal deaths

Figure 2

Risk Factor Prevalence in Live Births vs. Fetal Deaths (95% CI)



may reflect the same effect to a greater degree or may be attributable to intrauterine growth restriction.

The gestational age distribution for fetal deaths remaining flat across the entire pregnancy stands in stark contrast to live births, which show a rapid increase after week 32 peaking at week 39. This weight gap, combined with the absence of a distinct peak in the fetal death distribution, suggests that stillbirth risk factors may be present and impactful far earlier than the typical window for healthy delivery.

Figure 2: Risk Factor Prevalence

Figure 2 is a dot plot comparing the prevalence of selected maternal and pregnancy risk factors between live births and fetal deaths. Each row represents a single risk factor, with two dots indicating the proportion of records in each outcome group where that factor is present. Horizontal lines represent 95% confidence intervals; points without visible intervals indicate intervals smaller than the dot width due to the dataset's large size.

In most instances, risk factor prevalence is higher in the fetal death group. Many

factors show clear separation—for example, hypertension/eclampsia is present in less than 1% of records for both outcomes but is approximately three times as common in fetal deaths. Usage of fertility-enhancing drugs is the one variable with notably lower separation; while some fertility drugs are associated with increased stillbirth risk (Moore et al., 2025), the signal may not be strong enough in this dataset to meaningfully improve predictive performance.

Two factors break the expected pattern: gestational diabetes and gestational hypertension appear more prevalent in live births than fetal deaths. This likely reflects that these diagnoses trigger additional screening, enabling earlier identification and management of other complications. There is also a possibility that many fetal deaths occurred prior to when these conditions would have been diagnosed, skewing the comparison. These findings highlight the need for future modeling incorporating geographic access to healthcare data.

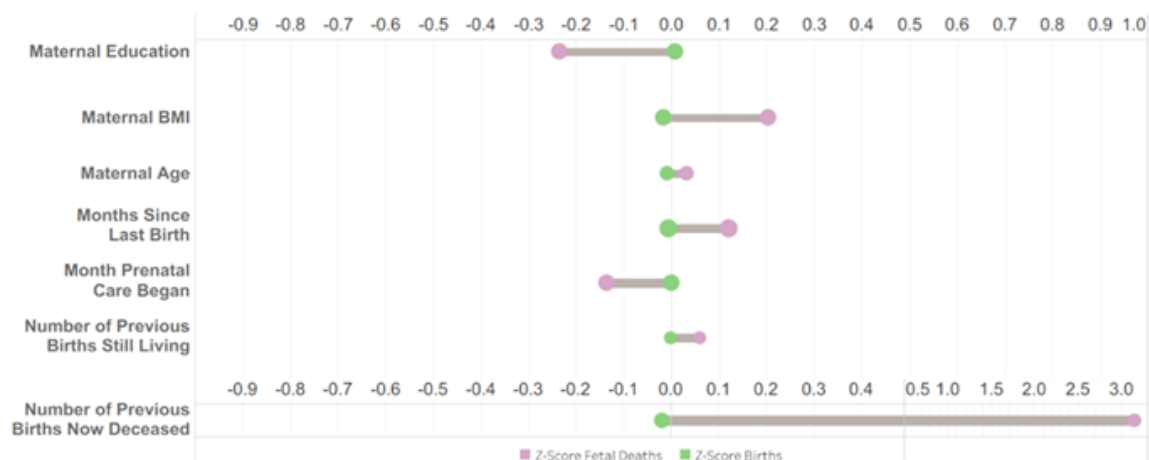
Figure 3: Standardized Mean Differences

The third visualization is a series of dumbbell charts. Each chart corresponds to a key maternal or pregnancy characteristic, with endpoints representing the Z-scored group mean for live births and fetal deaths respectively. Using Z-scores is necessary because these characteristics vary substantially in scale. The chart representing the number of previous live births now deceased uses a shifted x-axis scale to accommodate the extreme separation of that variable.

Several results aligned with prior expectations, including fetal deaths being associated with lower maternal education and higher maternal BMI. However, one surprising finding was that prenatal care began *earlier*, on average, in pregnancies that ended in fetal death—contrary to typical assumptions. Even after isolating for infertility treatment and other pre-pregnancy conditions, the trend persisted. This may suggest that these pregnancies were immediately identified as higher risk due to comorbidities, prompting earlier engagement with care.

The variable on previous live births now deceased is among the few that accounts for

Figure 3
Standardized Differences in Maternal and Pregnancy Characteristics by Outcome



prior adverse outcomes rather than characteristics of the current pregnancy. A mother with one or more prior infant deaths is more likely to have persistent risk drivers that do not disappear between pregnancies, such as chronic disease burden, barriers to consistent care, or environmental exposures. Additionally, sudden infant death syndrome (SIDS) is more prevalent among preterm or low-gestational-age infants (Rosenstein et al., 2013), meaning this variable may partially proxy a history of preterm labor, bundling multiple risk pathways into a single count.

Predictive Models

Multiple machine learning techniques were explored. Of those tested, the three strongest performing approaches were logistic regression, CatBoost, and AutoGluon.

Logistic Regression

The logistic regression classifier serves as a traditional baseline for binary classification tasks. It models the probability of a discrete outcome by mapping a linear combination of input features through the sigmoid function, relying on the assumption of a linear relationship

between the log-odds of the target and the independent variables (Stoltzfus, 2011). This model utilized an L2 penalty with a regularization strength (C) of 1.772. It produced an AUC of 0.748, supported by an overall recall of 0.799 and a precision of 0.895. For the minority class, the model achieved a sensitivity of 0.554, a precision of 0.211, and an F2 score of 0.418. The F2 score was used as the training optimization metric because it places greater weight on recall, which minimizes the higher cost of false negatives in a clinical context.

Feature importance for this model highlighted a strong reliance on data reporting nuances rather than traditional clinical indicators. The most influential variables were unknown status indicators for cigarette usage, infertility treatments, and various forms of hypertension. This suggests that the absence of a recorded value may serve as a proxy for specific patient profiles or the systematic reporting gaps discovered during EDA. More established risk factors, such as nulliparity, appeared later in the importance ranking. These findings underscore how many predictive signals capture how data is collected at least as much as what the underlying conditions predict.

CatBoost

Categorical Boosting (CatBoost) is a gradient boosting algorithm that addresses known limitations of other boosting implementations when handling categorical variables. Standard target-based encoding can cause overfitting, while one-hot encoding can create an unmanageable number of features. CatBoost uses *ordered boosting*, which processes data in a random sequence and ensures the model only learns from data points that appeared earlier in the sequence, preventing leakage (Prokhorenkova et al., 2019).

To address class imbalance, the model was trained with a `scale_pos_weight` of 11.5 and a learning rate of 0.1. Hyperparameters remained largely static across the different feature sets tested. The model achieved an AUC of 0.788, an overall recall of 0.840, and a precision of 0.903. For the positive class, it produced a recall of 0.572, a precision of 0.266,

and an F2 score of 0.465.

Qualitatively, the top predictors were the Boolean flag for a history of pregnancy loss, recoded maternal BMI, and the Boolean flag for unknown cigarette usage. The high importance of the history of loss variable aligns with prior research showing that prior adverse outcomes are among the strongest predictors of subsequent stillbirth (Khatibi et al., 2021). The prominence of the unknown cigarette usage flag echoes the logistic regression findings, potentially acting as a proxy for late-entry or inconsistent prenatal care. While the precision for fetal death remains modest at 0.266, a recall exceeding 57% indicates sufficient signal to assist clinicians in identifying pregnancies that warrant additional monitoring or intervention.

AutoGluon Ensemble

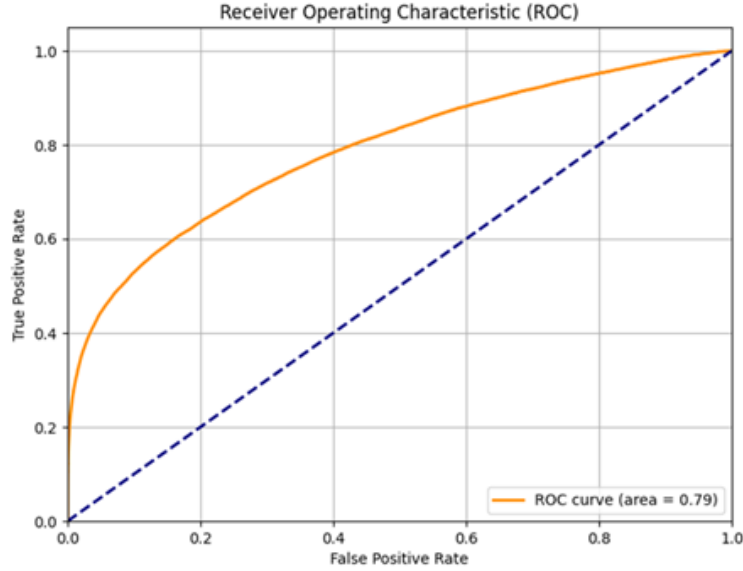
The AutoGluon model was by far the most complex architecture developed. AutoGluon is an AutoML framework that trains multiple models simultaneously and reduces the barrier to entry for machine learning while enabling best practices for advanced users (Erickson et al., 2020). The framework performs a model-agnostic data processing step followed by model-specific transformations, then stacks models across multiple layers while also passing the original data to higher levels—in a manner similar to skip connections in deep learning—enabling exceptional performance across most use cases (Erickson et al., 2020).

The final ensemble was composed of three layers. The third layer was a single weighted ensemble combining predictions from second-level models, which were mostly gradient boosting machines alongside a neural network and a weighted ensemble of first-layer models. The ensemble achieved the highest overall performance across nearly all metrics: an AUC of 0.790, an overall recall of 0.851, and a precision of 0.906. For the critical minority class, it reached a recall of 0.581 and a precision of 0.287, yielding a project-best F2 score of 0.482.

A notable distinction was AutoGluon’s prediction threshold of less than 0.1, compared to 0.38–0.64 for the other models. This likely reflects its ability to capture subtle signals in the minority class that simpler architectures overlook. The most important features were

Figure 4

AutoGluon ROC Curve ($AUC = 0.79$)



more closely aligned with traditional research findings: hypertension/eclampsia, number of prior fetal deaths, and maternal age ranked among the top predictors. By prioritizing these factors alongside a very low classification threshold, the model effectively shifts its decision boundary to favor sensitivity, ensuring that pregnancies with even marginal statistical risk are flagged for clinical review. Notably, the AutoGluon model ranked the age discrepancy between parents as the 11th most important feature, suggesting an ability to leverage proxies for socioeconomic information.

Model Comparison

Of the models trained, the AutoGluon ensemble achieved the highest performance across all primary metrics. Critically, its positive class F2 score was 0.0175 greater than that of CatBoost. While AutoGluon is the champion model for this study, the relatively narrow margin of victory suggests that the other architectures remain viable candidates for future research. The convergence of results across different architectures implies that the maximum predictive power is being extracted from the current dataset, and further gains would require

more granular data or advanced feature engineering techniques.

A significant finding was the general lack of consistency in feature importance across model architectures. Only four factors appeared in all three models’ top 20 features: BMI, pregnancy plurality, interval since last live birth, and the month prenatal care began. These variables likely represent the core physiological and socioeconomic drivers of fetal risk that are not sensitive to a given modeling approach. Outside of these core drivers, the top features were fragmented—the three models’ combined top-20 lists contained 45 unique values—despite AutoGluon being composed in part of a CatBoost model.

The models performed well relative to existing research, likely due to the application of modern ensemble techniques to a significantly larger dataset than previously studied. A systematic review by Li et al. (2025) provides a benchmark: the eight models analyzed reported an average AUC of 0.70 and recalls between 0.60 and 0.90, compared to this study’s champion AUC of 0.79. Notably, the top-performing model in that review (Khatibi et al., 2021) was also an ensemble; however, it included fetal weight and gestational age, which were excluded here as data leakage. This performance gap indicates that utilizing the full scale of CDC NVSS data provides a more stable foundation for risk prediction than smaller, localized samples.

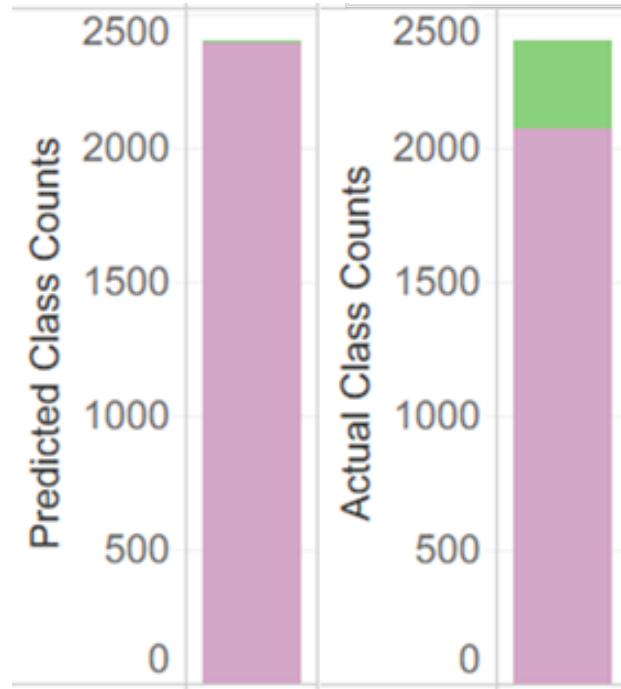
Results and Discussion

Findings

At the outset of the project, several systemic issues regarding the current state of ML fetal mortality research became apparent. Many studies fail to incorporate advanced modeling techniques or address significant data quality and processing concerns. Because fetal death is characterized by a complex multivariate environment where many risk factors are present even in uncomplicated pregnancies (Silver, 2007), advanced algorithms are well-suited to ingest the vast datasets created in the healthcare space to delineate structured risk levels.

Figure 5

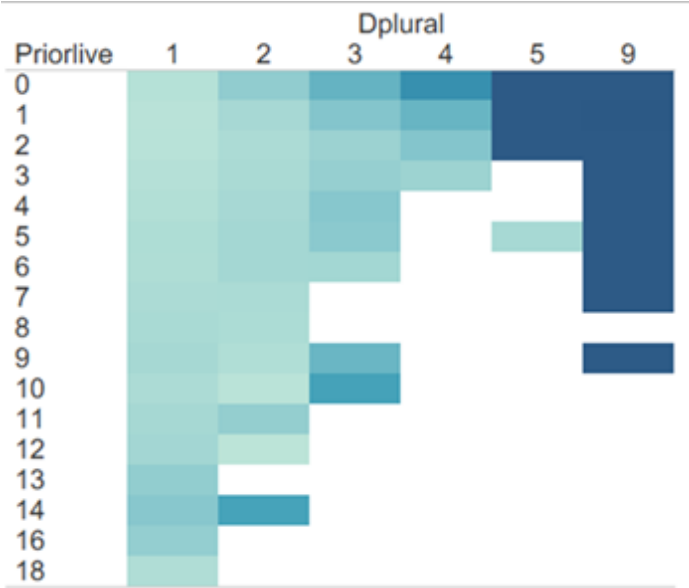
Predicted vs. Actual Class Distribution for Records with Six Unknown Risk Factors



Analysis of model performance relative to specific dataset features revealed a significant challenge regarding records with unknown data fields. As illustrated in Figure 5, the model failed to generalize records where all core risk factors were labeled as unknown; since 86% of such instances in the training set corresponded to fetal deaths, the algorithm consistently predicted a positive outcome for these cases with a high degree of confidence, reaching an average predicted probability of 90%.

Similarly, plurality emerged as the second most important feature, unexpectedly outranking all traditional risk factors. Closer inspection of the heatmap in Figure 6 revealed values of 1, 2, 3, 4, 5, and 9 for this feature. Value 9 is not a valid entry within any of the dataset's user guides; based on the coding patterns of other variables, it is clear that 9 is being used to represent unknown plurality. Critically, unknown plurality appears exclusively within fetal death records, allowing the model to learn that any record with unknown

Figure 6
Heatmap of AutoGluon’s Top Two Features (Prior Live Births $\times$ Plurality) vs. Predicted Probability

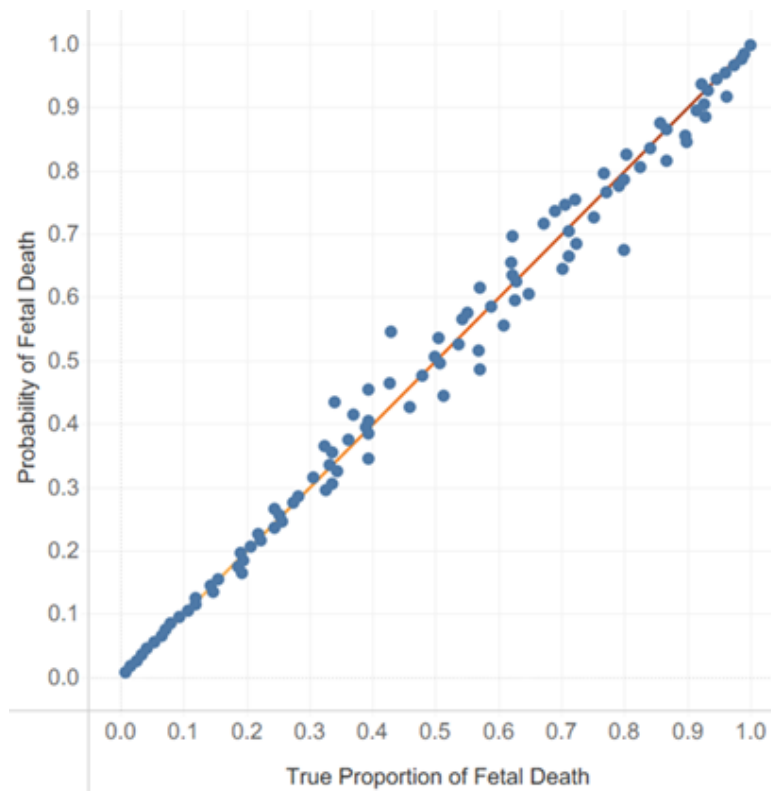


plurality is guaranteed to be a fetal death.

These observations align with the logistic regression model’s behavior, which also relied heavily on patterns in how data was collected rather than underlying clinical conditions. These findings indicate that systematic discrepancies between official documentation and the underlying data may lead to overstated model reliability and necessitate a more rigorous audit of how missingness is encoded across different birth outcomes. It is improbable, however, that these issues artificially inflate results to a degree that would alter key conclusions—the identification and removal of various data-leaking columns during modeling resulted in only a marginal impact on overall performance.

A key instrument for evaluating clinical utility is the calibration plot in Figure 7. This model demonstrates high probabilistic accuracy at lower risk thresholds. Between approximately the 40% and 75% thresholds, increased variability is observed—expected as certainty in either direction decreases. Above this range, the model trends slightly toward

Figure 7
Calibration Plot of the Final AutoGluon Model



overprediction of fetal death. Overall the model appears well-calibrated.

The confusion matrix in Figure 8 highlights the operational trade-offs inherent in the model’s configuration. By optimizing the F2 score, the model effectively minimized false negatives. In this evaluation group, false negatives represent the smallest of the four outcome groups—ideal for an application where failing to identify a high-risk pregnancy may result in an extremely adverse outcome. Although the model’s comparatively weak precision results in an 11.6% false positive rate, the substantial number of true positives relative to false negatives suggests that the model is successfully capturing a large proportion of total mortality events within the test set.

Figure 8*Confusion Matrix of Final AutoGluon Model*

Outcome	Predicted Outcome	
	0	1
0	367,013	52,833
1	15,289	21,228

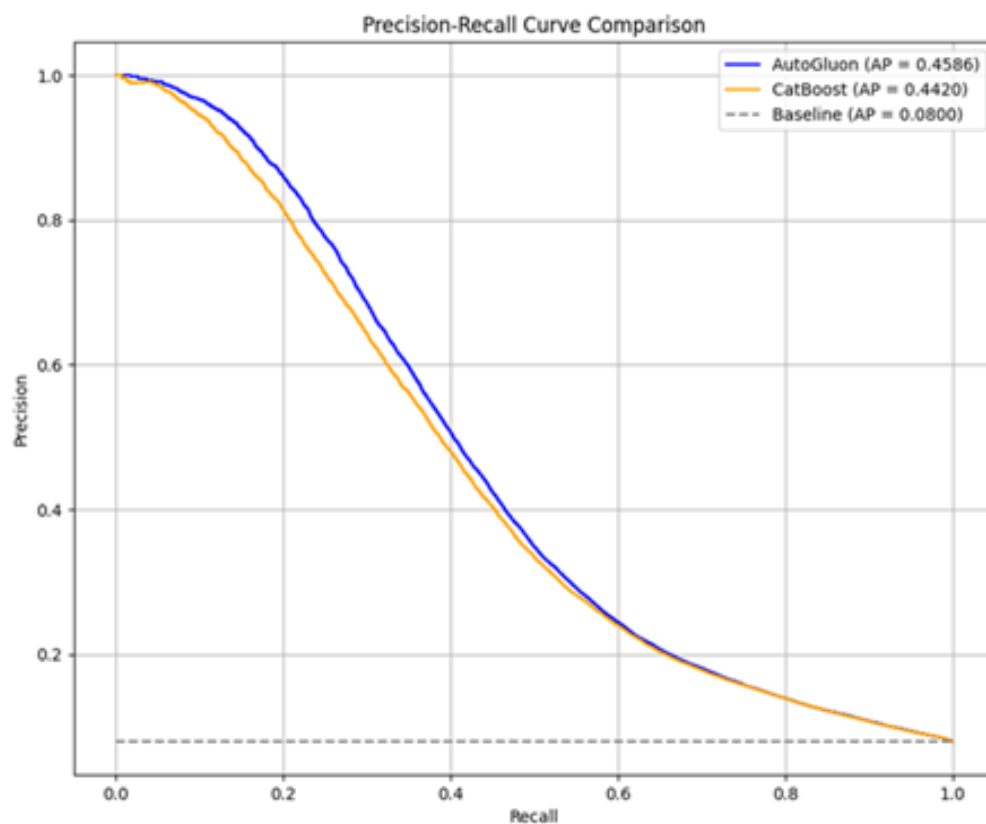
Performance Summary

The primary evaluation metric, the Precision-Recall curve, confirms a substantial improvement over the baseline. As illustrated in Figure 9, the baseline average precision for this imbalanced dataset is 0.08, whereas the final model achieved an average precision of 0.459. For the critical minority class, the final model achieved a recall of 0.581, while weighted recall across both classes reached 0.851. When compared to existing literature (Table 6), this model’s weighted precision of 91% represents an improvement over similar architectures applied to maternal health datasets.

As a further point of comparison, the Fetal Medicine Foundation has a widely recognized stillbirth risk calculator with a positive predictive value of less than 1% (Muin et al., 2022), compared to this study’s value of 29%. The AutoGluon model was successfully trained on consumer-grade hardware, demonstrating that significant predictive power does not necessarily require prohibitive computational resources.

Figure 9

Precision-Recall Curve Comparison: AutoGluon ($AP = 0.459$) vs. CatBoost ($AP = 0.442$) vs. Baseline ($AP = 0.080$)



Ethical Considerations

Fetal death is a particularly sensitive topic, underscoring the need for careful consideration of the ethical dimensions of this work.

Privacy

The Health Insurance Portability and Accountability Act (HIPAA) was designed to protect individuals’ protected health information (Public Health Law, 2024). The data used for this work includes all live births and fetal deaths within the United States, meaning individuals could plausibly identify themselves within it. However, with date information removed and no geographic information present, the introduction of machine learning has not added privacy concerns beyond those inherent in the public release of these files. Future work that incorporates the CDC’s restricted-access geographic data cannot be cleaned and uploaded to public repositories in the same manner, as a third party may be able to re-identify individuals—a HIPAA violation.

Explainability

An often-overlooked aspect of deploying machine learning in practice is model interpretability. Recently, the Artificial Intelligence Civil Rights Act was introduced, requiring developers to provide increased transparency around the use of algorithms in consequential decisions and granting individuals the right to appeal algorithmic decisions (Artificial Intelligence Civil Rights Act, 2024). The current model does not provide the level of decision-level insight required for independent clinical deployment. Any clinical implementation would require tools such as Shapley Additive Explanations (SHAP) or Local Interpretable Model-Agnostic Explanations (LIME) (Nori et al., 2019), alongside thorough clinician training on how to apply the model’s outputs as one input among many rather than a definitive determination.

Algorithmic Bias

Training data that reflects existing healthcare disparities may inadvertently perpetuate inequitable outcomes for marginalized populations unless it is handled properly. Because fetal death is an emotionally charged topic, the human element of care remains a necessary component of the clinical experience. Even in a scenario where a model’s predictive power is sufficient for independent operation, patients should retain access to a provider or clinical consultation tool for questions and concerns—ensuring that while backend calculations are driven by machine learning, the delivery of information remains accessible and supportive.

Conclusion and Future Work

Conclusion

The primary objective of this research was to determine whether the CDC’s NVSS data could be leveraged to develop an accurate ML model capable of predicting fetal death. The results suggest that such a model is not only feasible but also demonstrates robust performance across several critical metrics. The final model outperforms several contemporary models in the field, and the 29% positive predictive value compares favorably to the less-than-1% PPV of the Fetal Medicine Foundation’s widely used risk calculator (Muin et al., 2022).

Despite these results, several limitations warrant mention. No external validation was performed, so the model’s ability to generalize to international populations cannot be verified. While the EDA was extensive, it was not exhaustive—the plurality issue was not discovered until the modeling phase. Most significantly, because all variables are captured at the time of delivery or fetal death, the data reflects a state already shaped by medical interventions. This prevents establishing causality and limits the model’s utility as a real-time prognostic tool.

To facilitate further research and lower the barrier to entry for other investigators, the

curated dataset has been made publicly accessible on Kaggle¹ and the code is available on GitHub². Engagement metrics will be monitored over time to evaluate how the data science community receives this contribution.

Recommendations for Future Work

One of the most necessary avenues for future research involves external validation. Cabitza et al. (2021) argue that external validation is an indispensable requirement for ensuring the clinical reliability and generalizability of medical machine learning applications. Subsequent studies should seek to acquire analogous international datasets to determine whether the predictive patterns identified in the NVSS data persist across diverse healthcare systems and demographic profiles.

Future iterations must also transition from retrospective data toward time-phased datasets to mitigate the limitations of current reporting. Utilizing EHR data would enable inclusion of variables that change throughout pregnancy—ultrasound measurements, blood pressure readings, blood and urine analysis—providing a more dynamic predictive signal for identifying risk during rather than after a complication. Additionally, the model could be used to inspect the impact of various treatment strategies on a pregnancy’s projected risk, enabling a form of clinical scenario planning.

Finally, as the dataset grows in complexity, more sophisticated modeling architectures may be beneficial. Non-parametric transformers use an attention mechanism to inspect the entire dataset, effectively leveraging information from similar patient profiles to improve individual predictions (Kossen et al., 2021). While ensemble methods demonstrated strong value here, integrating prospective data and advanced architectures is the necessary next step toward translating these insights into clinical interventions that can reduce fetal mortality.

¹<https://www.kaggle.com/datasets/chrisheiner/national-vital-statistics-fetal-death-2016-2023>

²https://github.com/cheiner-27/fetal_death_prediction

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Appendix A

Data Profile Tables

Table A1

Neonatal and Natal Variables

Field	Description	Type	Blanks	% Blank	Outliers	% Outlier	Data Quality Issues
BWTR4	Birth Weight Recode 4	ordinal	0	0.00%	0	0.00%	
COMBGEST	Combined Gestation – Detail in Weeks	int	0	0.00%	391504	12.40%	Mixed Data Type
DBWT	Birth Weight – Detail in Grams (Edited)	int	0	0.00%	373891	11.84%	Data Leakage, Mixed Data Type, High Skew
DPLURAL	Plurality Recode	category	0	0.00%	17778	0.56%	
ME_PRE	Fetal Presentation at Delivery	category	7691	0.24%	0	0.00%	Data Leakage
OEGest_Comb	Obstetric Estimate Edited	int	0	0.00%	388351	12.30%	Mixed Data Type
OUTCOME	Live Birth or Fetal Death	boolean	0	0.00%	0	0.00%	
SEX	Sex of Infant	category	0	0.00%	0	0.00%	
YEAR	Year of Birth	int	0	0.00%	0	0.00%	

Table A2

Delivery Variables

Field	Description	Type	Blanks	% Blank	Outliers	% Outlier	Data Quality Issues
ATTEND	Attendant at Birth	category	0	0.00%	35068	1.11%	Data Leakage
BFACIL	Birth Place	category	7691	0.24%	64597	2.05%	Data Leakage
BFACIL3	Facility Recode	category	0	0.00%	868	0.03%	Data Leakage
DMETH_REC	Delivery Method Recode	category	0	0.00%	0	0.00%	Data Leakage
ME_ROUT	Final Route & Method of Delivery	category	7691	0.24%	15984	0.51%	Data Leakage
ME_TRIAL	Trial of Labor Attempted (if cesarean)	category	7691	0.24%	0	0.00%	Data Leakage
MM_AICU	Admit to Intensive Care	category	7691	0.24%	8756	0.28%	Data Leakage
MM_RUPT	Ruptured Uterus	category	7691	0.24%	2173	0.07%	Data Leakage
RDMETH_REC	Delivery Method Recode	category	7691	0.24%	21286	0.67%	Data Leakage

Table A3

Demographic Variables

Field	Description	Type	Blanks	% Blank	Outliers	% Outlier	Data Quality Issues
FAGECOMB	Father's Combined Age	int	0	0.00%	528848	16.75%	Mixed Data Type
FAGEREC11	Father's Age Recode 11	ordinal	0	0.00%	36036	1.14%	
FAGERPT_FLG	Father's Reported Age Used	boolean	3153235	99.84%	0	0.00%	Many Nuls
MAGER	Mother's Single Years of Age	int	0	0.00%	2353	0.07%	Mixed Data Type, High Skew
MAGER14	Mother's Age Recode 14	ordinal	0	0.00%	47602	1.51%	
MAGER9	Mother's Age Recode 9	ordinal	0	0.00%	11995	0.38%	
MAGE_IMPFLG	Mother's Age Imputed	boolean	3156032	99.93%	0	0.00%	Many Nuls
MBSTATE_REC	Mother's Nativity	category	0	0.00%	0	0.00%	
MEDUC	Mother's Education	category	7691	0.24%	0	0.00%	
MRACE15	Mother's Race Recode 15	category	7686	0.24%	100133	3.17%	
MRACE6	Mother's Race Recode 6	category	7686	0.24%	41597	1.32%	
MRACEHISP	Mother's Race/Hispanic Origin	category	7691	0.24%	30963	0.98%	
RESTATUS	Residence Status	category	0	0.00%	7404	0.23%	
WIC	WIC	category	7691	0.24%	0	0.00%	

Table A4

Maternal Health Variables

Field	Description	Type	Blanks	% Blank	Outliers	% Outlier	Data Quality Issues
BMI	Body Mass Index	float	7691	0.24%	230111	7.29%	Mixed Data Type
BMI_R	Body Mass Index Recode	ordinal	7691	0.24%	0	0.00%	Data Leakage
CIG_0	Cigarettes Before Pregnancy	int	7691	0.24%	330107	10.45%	Mixed Data Type, High Skew
CIG_1	Cigarettes 1st Trimester	int	7691	0.24%	278950	8.83%	Mixed Data Type, High Skew
CIG_2	Cigarettes 2nd Trimester	int	7691	0.24%	254203	8.05%	Mixed Data Type, High Skew
CIG_3	Cigarettes 3rd Trimester	int	7691	0.24%	372121	11.78%	Mixed Data Type, High Skew
CIG_REC	Cigarette Recode	category	7691	0.24%	0	0.00%	
DUMP_MM	Last Normal Menses Month	category	0	0.00%	198411	6.28%	
DUMP_YY	Last Normal Menses Year	int	0	0.00%	189944	6.01%	
ILLB_R	Interval Since Last Live Birth Recode	int	7691	0.24%	0	0.00%	Mixed Data Type, High Skew
ILLB_R11	Interval Since Last Live Birth Recode 11	ordinal	7691	0.24%	30270	0.96%	
IMP_SEX	Imputed Sex	boolean	3152240	99.81%	0	0.00%	Many Nuls
LBO_REC	Live Birth Order Recode	ordinal	0	0.00%	36939	1.17%	
M_HT_In	Mother's Height in Total Inches	int	7691	0.24%	127499	4.04%	Mixed Data Type
PRECARE	Month Prenatal Care Began	ordinal	7691	0.24%	18855	0.60%	
PRIORDEAD	Prior Births Now Dead	int	0	0.00%	173425	5.49%	Mixed Data Type, High Skew
PRIORLIVE	Prior Births Now Living	int	0	0.00%	133885	4.24%	Mixed Data Type, High Skew
PWgt_R	Pre-pregnancy Weight Recode	ordinal	7691	0.24%	1695093	53.67%	
RF_ARTEC	Asst. Reproductive Technology	category	7691	0.24%	19072	0.60%	
RF_EHYPER	Hypertension Eclampsia	category	7691	0.24%	8724	0.28%	
RF_FEDRG	Fertility Enhancing Drugs	category	7691	0.24%	24703	0.78%	
RF_GDIAB	Gestational Diabetes	category	7691	0.24%	0	0.00%	
RF_GHYPER	Gestational Hypertension	category	7691	0.24%	0	0.00%	
RF_INFTR	Infertility Treatment Used	category	7691	0.24%	0	0.00%	

Table A5

Reporting Flag Variables

Field	Description	Type	Blanks	% Blank	Outliers	% Outlier	Data Quality Issues
F_CIGS_0	Reporting Flag for Cigarettes before Pregnancy	boolean	907	0.03%	22034	0.70%	
F_CIGS_1	Reporting Flag for Cigarettes 1st Trimester	boolean	907	0.03%	22034	0.70%	
F_CIGS_2	Reporting Flag for Cigarettes 2nd Trimester	boolean	907	0.03%	22034	0.70%	
F_CIGS_3	Reporting Flag for Cigarettes 3rd Trimester	boolean	907	0.03%	22034	0.70%	
F_MEDUC	Reporting Flag for Education of Mother	boolean	907	0.03%	24980	0.79%	
F_MM_AICU	Reporting Flag for Admission to Intensive Care	boolean	907	0.03%	19900	0.63%	
F_MPCB	Reporting Flag for Month Prenatal Care Began	boolean	908	0.03%	29304	0.93%	
F_M_HT	Reporting Flag for Mother's Height	boolean	907	0.03%	21666	0.69%	
F_PWGT	Reporting Flag for Pre-pregnancy Weight	boolean	907	0.03%	23860	0.76%	
F_RF_GDIAB	Reporting Flag for Gestational Diabetes	boolean	907	0.03%	22111	0.70%	
F_RF_GHYPER	Reporting Flag for Gestational Hypertension	boolean	907	0.03%	22111	0.70%	
F_RF_PDIAB	Reporting Flag for Pre-pregnancy Diabetes	boolean	907	0.03%	22111	0.70%	
F_RF_PHYPER	Reporting Flag for Pre-pregnancy Hypertension	boolean	907	0.03%	22111	0.70%	
F_TOBACO	Reporting Flag for Tobacco use	boolean	907	0.03%	22034	0.70%	
F_WIC	Reporting Flag for WIC	boolean	907	0.03%	0	0.00%	
IMP_PLUR	Plurality Imputed	boolean	3155519	99.92%	0	0.00%	Many Nuls
MAGE_REPFLG	Reported Age of Mother Used Flag	boolean	3152264	99.81%	0	0.00%	Many Nuls
MRACEIMP	Mother's Race Imputed Flag	boolean	2909837	92.14%	198	0.01%	Many Nuls
OBGEST_FLG	Obstetric Estimate of Gestation Used Flag	boolean	2946283	93.29%	0	0.00%	Many Nuls